Deven Desai

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SUMMARY

Software Engineering Master's student and Stanford Hackathon winner specializing in AI agent workflows, real time data streaming, and scalable backend infrastructure. Demonstrated track record at Parspec and Terminal Trends engineering low latency retrieval pipelines, optimizing vector search performance, and building production agentic systems. Proficient in Python, C++, Go, TypeScript, Node.js, Vector Databases (Qdrant), RAG architectures, RESTful APIs, and asynchronous message queues.

PROJECTS

OmniTrack AI | Multimodal Agentic Engine & Real-Time Knowledge Pipeline GitHub

- Processed live streaming audio transcripts and unstructured metadata concurrently by engineering a zero to one AI agent backend using Python, Node.js, and Qdrant.
- Achieved sub 100ms vector retrieval updates by tuning indexing algorithms, vector embedding models, and parallel context synthesis workers.
- Reduced pipeline execution overhead by 30 percent by building automated streaming validation and fault tolerant retry mechanisms.

AgenticFlow | Real-Time Workflow Orchestration Platform GitHub

- Executed autonomous tool loops and multi step reasoning pipelines across cloud services by designing a low latency event orchestration backend using Go and Node.js.
- Increased backend throughput by 40 percent by implementing asynchronous WebSocket streaming interfaces and low overhead concurrent state locks.
- Maintained cross service data consistency by implementing automated integration tests and real time system monitoring.

Low Latency Search Engine | High-Throughput Indexing & Query System GitHub

- Queried and indexed millions of structured data records in real time by building a high performance analytical search core as a solo developer using C++ and Java.
- Reduced overall query execution latency by 35 percent through custom memory mapping, low level cache optimization, and efficient data structures.
- Enhanced search retrieval throughput by 20 percent by profiling thread contention and optimizing concurrent execution cycles.

WORK EXPERIENCE

Applied AI Intern, *Parspec, Inc., USA* **May 2026 – Dec 2026**

- Improved product-search relevance over a distributor inventory catalog, raising **recall@3 from 50% to 90%** on a **5,000-query** labeled evaluation set, by replacing keyword-only search with **hybrid** lexical and **semantic retrieval** fused via **reciprocal rank fusion**, re-scored with a **cross-encoder re-ranker**.
- Eliminated manual review of several-hundred-page construction documents for distributor bids, extracting structured product records at **88%** precision across **CSI Divisions 26 and 22**, by building a **RAG** pipeline over written specifications and schedules that grounds every extracted field in its source excerpt and auto-enriches contractor **Quantity Take-Off** line items.
- Automated lighting fixture product substitutions across 1.2M+ indexed datasheets at 85 percent match precision by engineering a retrieval pipeline that resolves free text inputs to anchor products and scores candidates via multi attribute filtering and semantic search.

Full Stack Developer, *Terminal Trends, India* **May 2024 – Dec 2024**

- Developed and optimized RESTful APIs using **Node.js** and **Express** to integrate real-time EV charging station data from multiple vendors (e.g., Tesla, government APIs), reducing server response time by **40%** and improving data reliability across the platform.
- Built interactive route-planning features with **React** and **Redux**, enabling users to locate nearby EV stations, view waiting times, pricing per kWh, and other live metrics along their journey, enhancing user convenience and engagement.

TECHNICAL SKILLS

AI & Agentic Systems: RAG Pipelines, Vector Databases (Qdrant), Autonomous LLM Agents, Knowledge Extraction, Contextual Retrieval, Prompt Engineering
Languages & Frameworks: Python, C++, Go, Java, TypeScript, Node.js, Express, React, FastAPI, RESTful APIs, GraphQL
Infrastructure & Data Systems: Low Latency Systems, Distributed Computing, Redis, PostgreSQL, Docker, Kubernetes, WebSockets, Event Driven Architecture
Software Engineering & Tools: Concurrency, System Design, Profiling & Optimization, Automated Testing, CI CD Pipelines, Git, Linux

EDUCATION

San Jose State University **San Jose, CA**
Master of Science in Software Engineering

ACHIEVEMENTS

- **Winner – Davis Wright Tremaine LLP Prize**, *Stanford Codex LLM x Law Hackathon (2025)*
  Led a 4-member team to build a **RAG-based legal QA engine** using Python, GPT-4, and vector-embedding retrieval with **cosine similarity**, enabling accurate extraction of contract clauses and delivering **90%+ correct answers**, outperforming 30+ teams.